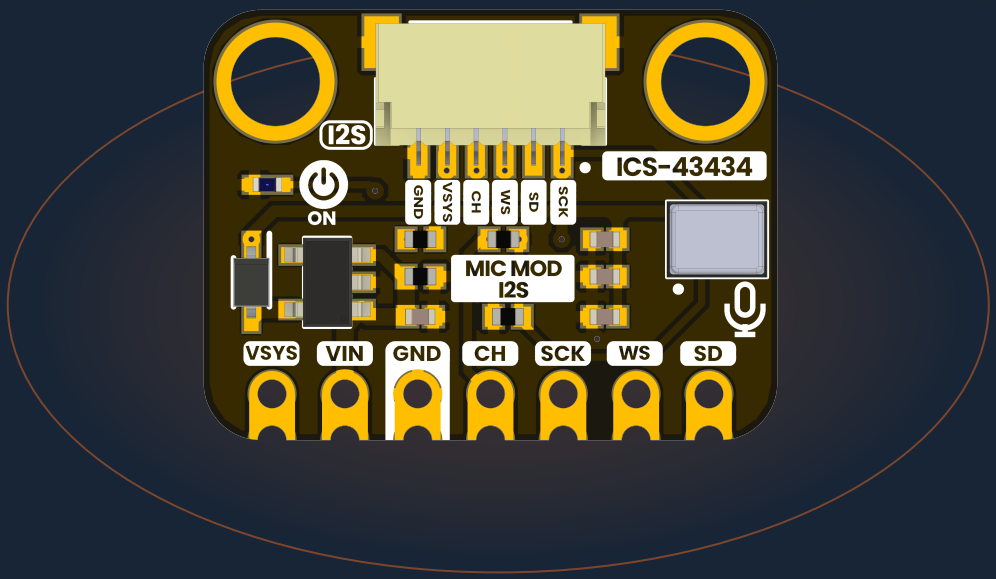
PRODUCT REFERENCE MANUAL

DEVLAB DIGITAL AUDIO MODULE

DevLab I2S ICS-43434 MEMS Microphone

MPN: UE0149

Version 0.1.0



Compact ICS-43434 MEMS microphone module with 24-bit I²S audio output, selectable channel assignment, and onboard 3.3 V regulation.

I²S AUDIO | 24-BIT OUTPUT | CHANNEL SELECT | MEMS

DOCUMENT MAP

Contents

Description
Applications
Hardware Features

1 The Module
1.1 Board Identification
1.2 Interface Hardware

2 Electrical Characteristics
2.1 Electrical Values
2.2 Electrical Precautions

3 Functional Overview
3.1 Functional Block Diagram
3.2 Signal Path
3.3 I²S Frame and Channel
3.4 Sample-Rate Modes
3.5 Power and Board Circuit

4 Connectors and Pinout
4.1 Edge Connections
4.2 Six-Position JST-SH Connector
4.3 Solder Options

5 Board Operation
5.1 Wiring and First Power-Up
5.2 Arduino Library Example
5.3 Other Capture Examples and Checks

6 Mechanical Information
6.1 Top View
6.2 Bottom View
6.3 Board Dimensions

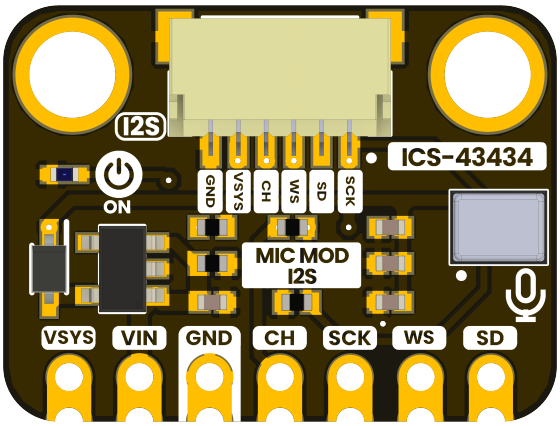
7 Company Information

8 Reference Documentation

9 Appendix
9.1 Technical Basis
9.2 Document Scope
9.3 Document Control

Description

The UNIT DevLab I²S ICS-43434 MEMS Microphone is a compact digital-audio module based on the bottom-port ICS-43434 MEMS microphone. The microphone integrates signal conditioning, a 24-bit analog-to-digital converter, decimation and anti-aliasing filters, power management, and an I²S output.



Applications

- Voice capture and recognition prototypes
- Intercom and teleconferencing systems
- Acoustic sensing and security projects
- Stereo or microphone-array experiments
- Educational I²S audio acquisition

Hardware Features

- ICS-43434 bottom-port digital MEMS microphone
- 24-bit I²S serial audio output
- Left/right time-slot selection through the CH signal
- AP2112K fixed 3.3 V onboard regulator
- Seven 2.54 mm edge connections
- Six-position, 1.0 mm-pitch JST-SH connector
- Power indicator and selectable solder options shown on the V1.0 artwork

1 The Module

1.1 Board Identification

Item	Value
Product	DevLab: I2S ICS-43434 MEMS Microphone
Manufacturer part number	UE0149
Hardware revision shown	V1.0
Microphone	ICS-43434
Digital audio interface	I^2S
Product-reference revision	0.1.0

1.2 Interface Hardware

The module provides seven 2.54 mm edge connections and a six-position, 1.0 mm-pitch JST-SH connector. The connector carries `GND`, `VSYS`, `CH`, `WS`, `SD`, and `SCK`. Its signal order follows the board pinout in Section 4.

2 Electrical Characteristics

2.1 Electrical Values

Symbol / parameter	Description	Value	Scope
VDD	ICS-43434 microphone supply	1.65 to 3.63 V	Microphone pin, DS-000069 rev. 1.2
ADC output	Digital sample width	24 bit	Microphone I^2S output
SNR	Signal-to-noise ratio	65 dBA high-performance; 64 dBA low-power, typical	Microphone characteristic
Frequency response	Stated audio band	60 Hz to 20 kHz	EMMIC-ICS43434 reference
Sensitivity tolerance	Unit-to-unit tolerance	±1 dB	Microphone characteristic
VREG	AP2112K-3.3 regulated output	3.3 V nominal	Module regulator

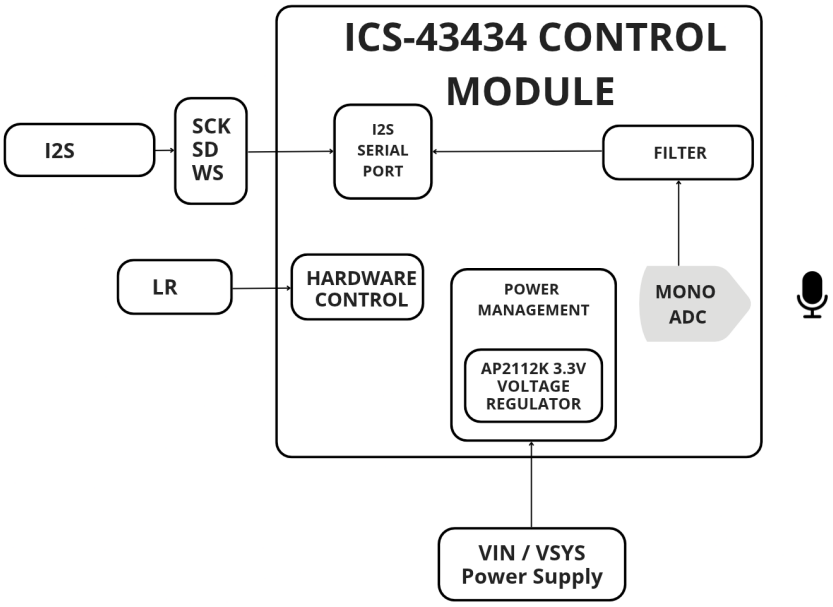
The sensor VDD range applies at the microphone supply pin. The module routes VIN through its regulator and Schottky diode to VSYS, which supplies the microphone. The sensor VDD range does not specify a module VIN input range.

2.2 Electrical Precautions

- The host and module share a common ground during operation.
- The host drives SCK, WS, and CH within the powered microphone's logic domain.
- The microphone drives SD toward the host.
- The bottom acoustic port remains clear of debris, flux, adhesive, and enclosure pressure.
- The board schematic shows the VIN to VSYS supply path and the JST power connection.

3 Functional Overview

3.1 Functional Block Diagram



UE0149 I²S ICS-43434 microphone module block diagram.

The diagram summarizes the microphone signal path, channel control, and board power section. The host drives SCK and WS toward the microphone; SD carries audio from the microphone to the host. Section 2 specifies the sensor supply range.

3.2 Signal Path

Sound enters the ICS-43434 through its bottom acoustic port. The MEMS signal passes through conditioning, an ADC, decimation and anti-aliasing filters, then leaves as signed 24-bit, two's-complement I²S data on SD. The host supplies SCK (bit clock) and WS (word select); the sensor is an I²S slave and needs no separate master-clock pin. The manufacturer datasheet, DS-000069 revision 1.2, describes this path in "Theory of Operation."

3.3 I²S Frame and Channel

Each stereo frame contains exactly 64 SCK cycles: 32 per channel. The sensor transmits 24 valid bits, most significant bit first, starting one bit-clock after the WS transition. A host normally receives 32-bit slots and converts the signed sample to its application format. Board CH reaches the microphone LR input: low selects the left slot after the falling WS edge, and high selects the right slot after the rising edge. The host firmware receives the selected slot.

The microphone tri-states SD after its 24th bit. A stereo pair shares SCK, WS, and SD when one microphone selects left and the other right. For that arrangement, the datasheet specifies a pull-down on the shared SD line (100 kΩ is sufficient). The UE0149 circuit includes a 10 kΩ pull-down on CH, so the default channel is **left**. The assembled solder option determines the effective selection.

3.4 Sample-Rate Modes

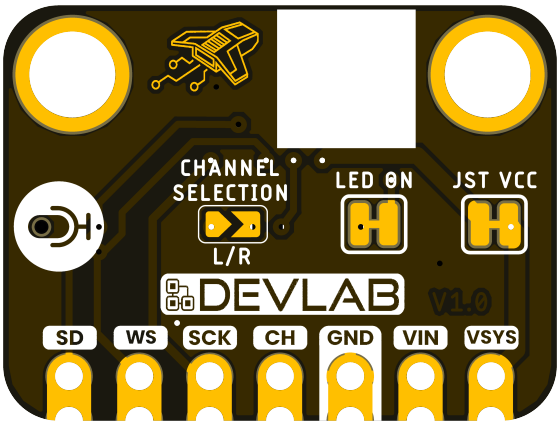
The sensor selects its power state from WS frequency (the sample rate):

Sensor mode	WS frequency	Typical sensor current	Typical SNR
Standby/sleep	Below 3.125 kHz	12 µA	No data output
Low power	6.25–18.75 kHz	230 µA	64 dBA
High performance	23–51.6 kHz	490 µA	65 dBA

The table lists **sensor** characteristics from DS-000069. Rates between the listed ranges lie outside the specified operating modes. A 48 kHz, 64-clock frame uses a 3.072 MHz bit clock and falls in high-performance mode. The microphone takes up to 20 ms to reach settled output after power-up or wake-up. In standby, the host holds both SCK and WS low.

3.5 Power and Board Circuit

The circuit routes VIN through an AP2112K-3.3V regulator and a Schottky diode toward VSYS, which powers the microphone VDD pin. A 100 nF capacitor decouples the microphone, and the board has an orange power LED. The ICS-43434 VDD operating range of 1.65–3.63 V applies at the sensor pin; the module supply path determines the voltage at that pin. The host holds I²S clocks low while the sensor is unpowered.



4 Connectors and Pinout

4.1 Edge Connections

Board label	Direction	Function
VSYS	Power	Module system rail
VIN	Input	Regulator input
GND	Power	Common ground
CH	Input	Left/right I²S channel selection
SCK	Input	I²S serial bit clock
WS	Input	I²S word select / left-right clock
SD	Output	I²S serial audio data

4.2 Six-Position JST-SH Connector

The V1.0 top artwork labels the connector signals, from the leftmost contact in the rendered top view, as GND, VSYS, CH, WS, SD, and SCK. The board pinout shows signal order by physical position.

4.3 Solder Options

The bottom artwork identifies L/R, LED ON, and JST VCC options. The circuit pulls CH low through 10 kΩ and selects the left channel by default. The assembled solder bridges determine channel selection, LED operation, and JST supply routing.

5 Board Operation

This example uses the DevLab_AudioInput library to capture the ICS-43434's I²S output as signed 16-bit mono PCM on an ESP32-S3. The library supports ESP32-S3 hardware capture and compiles for RP2040/RP2350.

5.1 Wiring and First Power-Up

The connection uses a clear acoustic port, a common ground, and a supply that follows the module power circuit. The board revision and solder options determine the effective channel selection. The sensor VDD range describes the microphone pin rather than the module VIN input.

UE0149 signal	Example ESP32-S3 pin	Direction
SCK	GPIO36	Host I²S bit clock output
WS	GPIO37	Host I²S word-select output
SD	GPIO35	Host I²S data input
CH	Board default low	Left time slot
GND	GND	Common ground
VIN or VSYS	Supply matching the board power circuit	Board power

The UE0149 circuit pulls CH low and selects the left slot by default. If the assembled board selects right, the sketch uses I2S_STD_SLOT_RIGHT. The GPIO numbers identify host pins. The host board determines which GPIOs are available for the example.

5.2 Arduino Library Example

The example requires the library in the Arduino sketchbook and an ESP32-S3 with Arduino-ESP32 3.x. The complete AudioInputBasicRead sketch reports the absolute peak of each PCM16 block at 115200 baud:

```
#include <DevLab_AudioInput.h>
DevLab_AudioInput microphone;
int16_t samples[256];

void setup() {
  Serial.begin(115200);
  if (!microphone.beginI2S(GPIO_NUM_36, GPIO_NUM_37, GPIO_NUM_35,
                           48000, I2S_STD_SLOT_LEFT)) {
    Serial.println("I2S initialization failed");
    while (true) delay(1000);
  }
  delay(25);
}

void loop() {
  size_t count = microphone.read(samples, 256);
  if (count == 0) return;
  int32_t peak = 0;
  for (size_t i = 0; i < count; ++i) {
    int32_t value = samples[i];
    if (value < 0) value = -value;
    if (value > peak) peak = value;
  }
  Serial.println(peak);
}
```

The Serial Monitor displays the peak at 115200 baud. The peak changes with sound level and represents digital PCM amplitude. At 48 kHz, a logic analyzer shows a 3.072 MHz SCK and 64 bit clocks per WS frame.

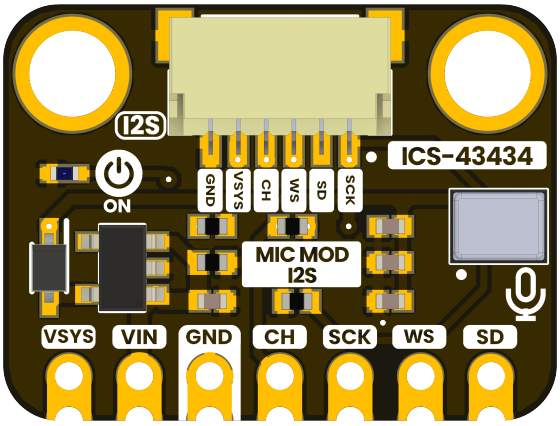
5.3 Other Capture Examples and Checks

The library's `examples/I2S` folder also provides `SerialPlotter`, `SerialStream`, and `ComputerAudio`. Its stock examples drive their selection pin high and receive the right slot. The UE0149 default selects the left slot, so the application sets CH high or receives the left slot. The existing LegacySerialPlotter sketch uses the older ESP32 I²S driver directly. It selects the board's drawn left slot and is retained as an experimental legacy sketch.

If initialization fails, the host configuration checks the Arduino core version and GPIO availability. If samples remain near zero, the signal check covers module power, SCK, WS, SD, and the selected slot relative to CH. If output clips or is noisy, the check covers the conversion shift, supply, wiring, and acoustic port.

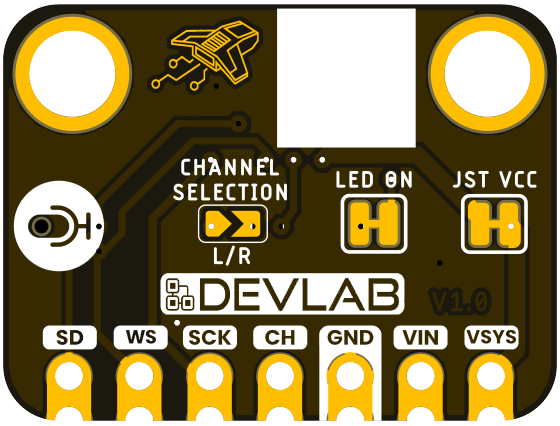
6 Mechanical Information

6.1 Top View



UE0149 I2S microphone top view.

6.2 Bottom View



UE0149 I2S microphone bottom view.

The bottom acoustic port remains clear of adhesive, contamination, and mounting hardware.

6.3 Board Dimensions

The dimension drawing is stored with the board images.

The labeled drawing provides the board dimensions for mechanical layout. The physical assembly determines connector clearance and manufacturing tolerances.

7 Company Information

Item	Information
Company	UNIT Electronics
Website	https://uelectronics.com/
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The repository in Section 8 provides the current product documentation.

8 Reference Documentation

Reference	Location
Repository	UNIT-Electronics-MX/unit_devlab_i2s_ics_43434_mems_microphone
Hardware notes	Hardware README
Pinout PDFs	English · Spanish
Pinout images	English PNG · Spanish PNG
Microphone reference	EMMIC-ICS43434 PDF
Manufacturer sensor datasheet	DS-000069 rev. 1.2
Module schematic	UE0149 schematic
Capture library	DevLab_AudioInput
Module block diagram	UE0149 block diagram
Library-based example	UE0149 AudioInputBasicRead
Experimental firmware	Legacy I²S example
Top view	3D board view
Bottom view	3D board view
CAD model	STEP

The manufacturer document DS-000069 revision 1.2 (2016-08-29) supplies the sensor timing, modes, and interface characteristics. The UE0149 schematic supplies the module circuit connections, and the board artwork supplies the visible signal labels. The EMMIC-ICS43434 file describes a separate breakout board.

9 Appendix

9.1 Technical Basis

This product reference follows the V1.0 board artwork for visible signal labels, the UE0149 schematic for circuit connections, and manufacturer datasheet DS-000069 revision 1.2 for ICS-43434 characteristics.

9.2 Document Scope

- Section 2 lists the ICS-43434 sensor characteristics and the module's nominal regulated voltage.
- Sections 3 and 4 describe the signal path, power circuit, and board connections.
- Section 5 shows an ESP32-S3 capture example using the DevLab_AudioInput library.
- Section 6 shows the board views and dimension drawing.

9.3 Document Control

Field	Value
Product	DevLab: I2S ICS-43434 MEMS Microphone
Manufacturer part number	UE0149
Hardware artwork covered	V1.0
Product-reference revision	0.1.0
Publication date	2026-08-03
Firmware example	ESP32-S3 I^2S capture in Section 5